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Securing wireless sensor networks: A survey of challenges and innovations

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ABSTRACT

WSNs are a significant component of the modern technology systems. They find application in numerous sectors such as environmental monitoring, health care, agriculture and intelligent constructions. Although they possess low power, have low processing capacity and can readily be influenced by unsafe environments, they are still very important. They are therefore usually the victims of various forms of security threats as a result of that. The more advanced these threats become, the higher is the need to have strong and flexible security measures. The survey examines a broad spectrum of studies on the security of WSN. It encompasses numerous fields, such as how to transmit data safely, how to deal with keys, trust and reputation management systems and how to check and authenticate users. It also considers novel concepts such as applying blockchain and machine learning to identify security problems. Some of the studies in the research are based on straightforward solutions that can be applied effectively with the available limited resources of WSNs, whereas others attempt to achieve the balance between security and efficiency. Despite these developments, there are issues. The solutions are mostly energy consuming or are rigid to adapt to changing network conditions. The disconnect between the concepts in studies and their practical implementation is also present, in particular, the use of complex technologies, such as AI, in practice. This review aims to unite significant conclusions of the recent research, to emphasize the main issues that are not resolved yet, and to propose the useful directions in the further investigations. This survey provides an up-to-date and clear image of the current trends in WSN security, as it allows the researcher and developers to develop more efficient and profound strategies on how to make the system stronger and more reliable.

1. Introduction

WSNs are significant in intelligent systems which interact with the physical world. They aids in various fields such as animal tracking, the health of buildings, making factories more efficient, healthcare services, and emergency control. They are good since they can grow easily, get adapted to various situations and are not too costly. But WSNs operate in environments where they possess minimal power, memory, internet speed and computing capacity. They are vulnerable to a lot of attack being left alone as they use wireless connections and most of the time when they are alone.

Fig. 1 depicts the arrangement of WSNs in a layered fashion. The tiny sensors referred to as sensor nodes transmit data to cluster heads followed by a main station and eventually alocation that oversees the data remotely. Such arrangement simplifies the system to manage, conserves energy and aids in understanding the information at a single site. The safety of WSNs is highly paramount since when they are compromised, this may result in misleading information, individuals who spy and make poor decisions. Some of the threats that are likely to occur are a person taking over a device, altering the information, eavesdropping, replaying messages, impersonating multiple devices, or halting the functioning of the system.

Fig. 2 indicates the various security threats in WSNs, categorized according to the network layers. Beginning with the physical layer and up through the application layer, with risks that may be experienced by more than one layer at a time. The jamming which blocks signals, node capture which occupies devices, tampering and eavesdropping where messages are overheard threatens the physical layer. These may result in misunderstanding and lead to a failure of communication and endangering sensitive data. The data link layer is prone to collision attacks (messages are mixed up), exhaustion attacks (messages use resources), and MAC spoofing (phony devices impersonate real ones). These attacks have the ability to damage the reliability of the network besides determining who can use the network. Network layer is vulnerable to attacks such as sinkhole where the attacker uses deceptive nodes to influence other nodes into transmitting data, wormhole where messages are

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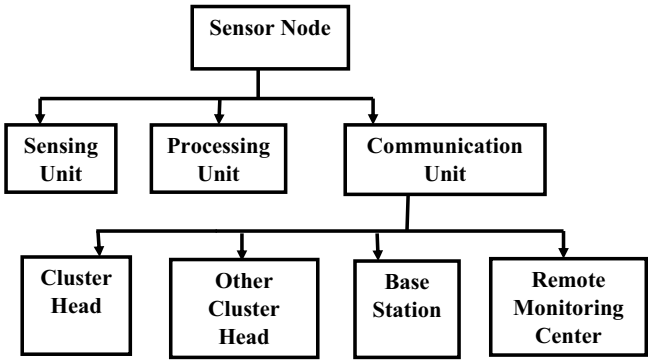


Fig. 1. General Architecture of a WSN.

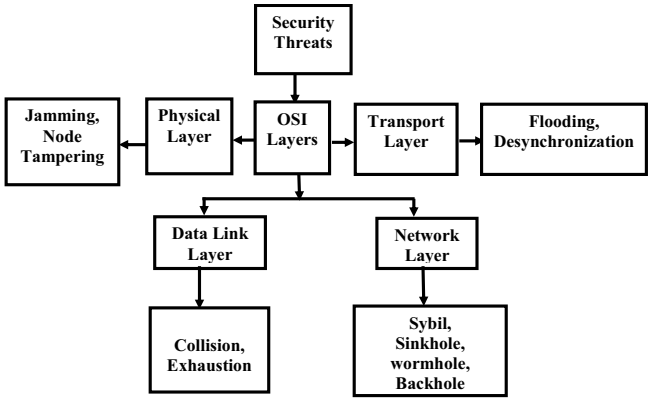


Fig. 2. Classification of WSN Security Threats.

transported under cover to remote points and Sybil where the attacker uses fake nodes to take control of the network. These have the ability to alter the routing of data and network design. Flooding, excessive transmission of data, and desynchronization, respectively, jeopardize the transport layer, when overabundance of data is transmitted, and timing problems arise.

Data integrity problems can be encountered in the application layer, with some information being modified, malicious code being introduced into the system, and privacy breaches being experienced by revealing personal information. Cross-layer attacks exploit several layers simultaneously e.g. energy depletion where the network exhausts its power supply, and adaptive attacks where the network adapts and learns against already implemented defenses. Threats of WSNs can be classified by layer so that it is easy to develop layer-specific security solutions, allocate resources effectively, and develop robust inter-layer mechanisms that operate effectively.

WSNs have now become applicable in such fields as healthcare, environmental surveillance and smart cities. However, due to their low power, as well as processing capability, they are simpler to attack. The conventional encryption tools tend to consume a lot of power and resources that these networks cannot accommodate and hence efficient and lightweight security tools are required. To remedy this, the study recommends a safe routing protocol, which integrates energy efficiency, trust-sensitive framework, and Attribute-Based Encryption (ABE) to ensure security of data and to make the network more efficient. Current techniques are usually too powerful and lack includes trust measures, which demonstrates the necessity of a comprehensive solution such as this [40].

The multilayer security system created on WSNs is shown in Fig. 3. It encompasses all the levels of the OSI model, which are all the ways of detecting tampering at the lowest level, which is the physical layer, and the processing of data safely at the highest level, which is the application

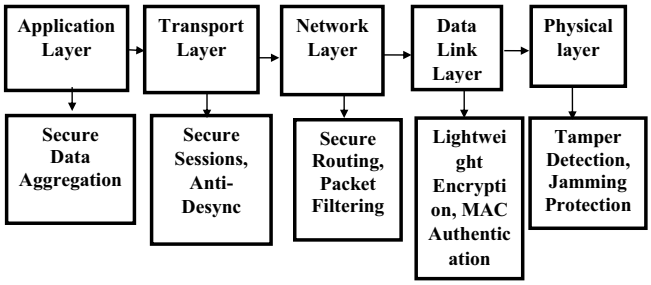


Fig. 3. The idea of a multilayer approach of WSNs emphasizes the interactions between the various security mechanisms in the physical, network, and application layers.

layer. It acts as a shield of protection, and every layer has its degree of protection. The framework consists of five layers, including Physical, Data Link, Network, Transport and Application (Fig. 4).

Collectively, they assist in maintaining a smooth running communication, effectively route data, and also assist in supporting different kinds of applications. One of the things that can be identified is the fact that the framework does not consider each layer distinctly. Rather, it has strategies that unify the layers and this saves energy and enhances the performance of the network. The other desirable aspect is the associations between the layers.

In case something goes wrong in one of the layers such as a sharp jamming signal or unusual activity can be reported to the rest. This enables the network to adapt rapidly and make security measures where necessary. It is not a dogmatic system, consequently, it can change, and respond to the events unfolding in practice.

The framework can also adjust its security settings to the requirements of the factors including the level of threat, the roles being played by the respective node and the energy available after the consumption. This is useful especially in WSNs where there is never abundance of resources available. In general, this architecture proposes a scalable and flexible architecture to a range of WSN applications, such as management of critical infrastructure or the environment. It also

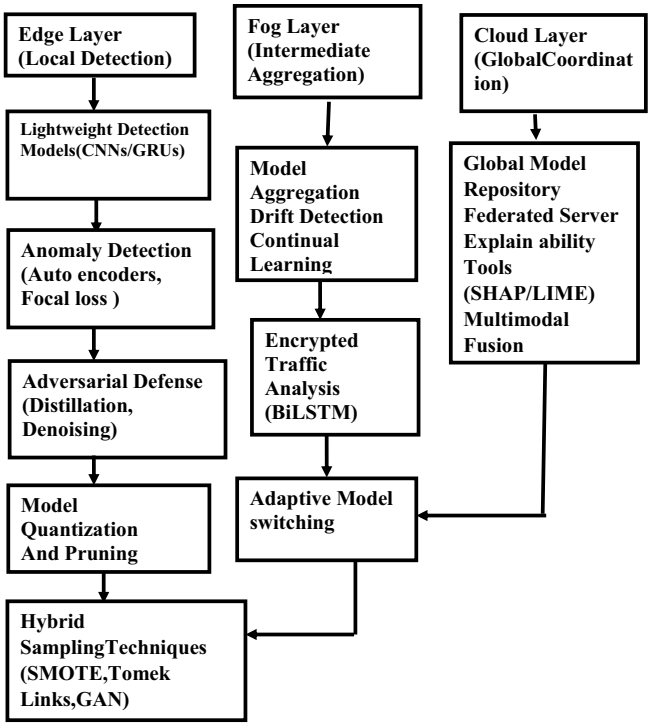


Fig. 4. Representation of the AIDE-FL architecture.

increases the network to be more resistant to complex attacks. It introduces additional functionalities like intrusion detection, trust-assessment, and safe routing on more than one layer and constitutes a strong and secure network which has lesser chances of being attacked.

The methods of solving security problems have been innumerable and have been practiced over the years. Other people have developed crude rules of authentication of persons who ought to be permitted to use a system as well as privacy of messages.Others have learned special techniques of dealing with keys under circumstances where things are always evolving and scattered. It has also been proposed by many people how to send messages safely, realize when something is going wrong and how to cope with situations when connections are lost or some part of the system is not functioning correctly. Although there have been numerous good ideas there has not been a one-solution that fits in every circumstance.

The work pays attention to the latest advancement in the area of WSN security, such as distributed trust models, lightweight cryptography, and machine learning-based intrusion detection. The study presents a more detailed comparison as opposed to the earlier surveys, which only present the general picture, and raise the issue of trade-offs in performance, energy consumption, and scalability. It further categorizes the approaches that exist under the various security aspects. In addition to the summary of earlier research, this survey highlights the gaps in the research that are open to examination and the directions of future investigation which encompass cross-layer security models, federated learning, and adaptive, power-efficient WSNs.

Furthermore, the paper evaluates some of the most important research articles that have contributed to the sphere greatly [11]. The areas of these works include cryptographic tools, design of secure communication, methods of detecting intrusion, and models of trust. This is based on the analysis of their key contributions, methodologies, tools or simulation platforms used. The discussion of their strengths and weaknesses in order to bring out the synthesis of an extant, structured model of where the society is today and where it should be going next. The need to create more intelligent, more adaptive, and more power-efficient security mechanisms in the WSNs has never been as strong through the prism of machine learning, distributed trust, lightweight security protocols, or cross-layer designs.

Prior to the methodological and architectural sections, it should be mentioned that the discussed AIDE-FL framework is a conceptual one. It is not an established or tested system. The framework will help to unite a number of developed methods into a single framework. They are federated learning, constant model updating, multimodal data integration, and adversarial manipulation defenses. The concept is to develop a scalable framework that is capable of supporting contemporary intrusion detection requirements. The prime goal is to provide a systematic roadmap towards approach to various critical issues in intrusion detection. These obstacles include real-time flexibility, privacy of user data, decryption of encrypted data, and low resource operation. Later some preliminary experimental work is given. These findings indicate that the proposed model is useful and possible. Using this conceptualization of AIDE-FL, the study provides a solid basis to the further research. It also provides the opportunities to make a practical implementation in dynamic and complex network setups.

2. Literature review categorization and comparison framework

2.1. Security challenges in WSNs and IoT

WSNs and IoT networks are susceptible to diverse security threats, including Denial of service attacks, routing forgery, data alterations, and privacy. The low power consumption, information from various devices, and concerns about security from large networks are the challenges being addressed.For instance, Rohman et al. (2023) [14] have introduced a detailed work on energy modeling with link layer protocols in WSNs with MinyOS and TinySec. It married the Low-Energy Adaptive Clustering Hierarchy (LEACH) protocol with TinySec to improve energy efficiency and security. It was found through the study that LEACH protocol with the use of the light encryption of TinySec was able to decrease energy consumption considerably while keeping the confidentiality and integrity of the data. The study therefore provides useful exposure for energy-efficient protocol development having secured data transmission in WSNs, and is an important foundation for future studies in this area.

2.2. AI-based intrusion detection systems

2.2.1. Machine learning approaches

Aruchamy et al. (2023) proposed an AI-enabled, energy-efficient intrusion detection and secure routing framework for IoT-enabled WSNs. The authors report 95 % intrusion detection accuracy and energy savings of 38 % over baseline protocols. The exact dataset studied is not explicitly stated, but the paper explains how modern efforts expand baseline WSN security paradigms such as TinySec and SPINS [8].

Seth et al. designed an IDS model with a hybrid feature selection approach and Light Gradient Boosting Machine (LightGBM), which decreased prediction latency by 44.52 % and model development time by 52.68 %. Their method obtained 97.73 % accuracy, 96 % sensitivity, and 99.3 % precision [41].

Ahmad, R., Wazirali, R., & Abu-Ain, T. suggested a t-SNERF-based model for dimensionality reduction and feature correlation, trained on RF. The proposed model had 100 % accuracy on UNSW-NB15, 99.70 % on Phishing, and 99.78 % on CIC-IDS2017 datasets.

Talukder(2024) offer MLSTL-WSN, it is based on SMOTE-Tomek for resolving data imbalance. By using a wide WSN dataset, it is going to a range of machine learning models to provide over 99 % accuracy in multiclass as well as binary detection. The model is effective and optimized for WSN environments with limited resources, providing security via effective detection of threats [2].

2.2.2. Deep learning approaches

Ashiku, L., &Dagli, C assessed the DL model performance of DNN, RNN, and ANN for IDS binary and multiclass tasks using the UNSW-NB15 dataset. Preprocessing data with cleaning and normalizing techniques, they achieved accuracies of 99.26 %, 85.42 %, and 99.22 % on binary classification and 97.89 %, 85.4 %, and 95.9 % on multiclass tasks respectively [42].

Ahmad, Z., developed a DNN-based IDS for IoT attacks. They evaluated their model on UNSW-NB15, KDDCUP'99, and NSL-KDD datasets with an overall accuracy of 91.5 % on all datasets [43].

Godala, S., &Vaddella, R. P. V. Presented a hybrid model of IDS based on LSTM and CNN supplemented with STL (SMOTE + TomekLink) for the management of class imbalance. On PySpark with CICIDS-001 and UNSW-NB15 datasets, the model yielded 99.83 % multiclass and 99.17 % binary classification accuracy [44].

Madhuri, K. et al. emphasized the use of CNN to detect DoS attacks in comparison with RNN using KDDCUP'99 and CSE-CIC-IDS2018 datasets. CNN showed a value over 99 % both on binary and multiclass on KDDCUP'99 and 91.5 % on CSE-CIC-IDS2018 while being better than RNN that attained 99 % and 93 % on KDDCUP'99 and 65 % on the latter.

2.2.3. Hybrid approaches

Krishnan, R et al.proposed a hybrid model of intrusion detection by combining Deep Neural Networks (DNN) with Ant Colony Optimization (ACO) to optimize hyper parameters for maximum performance. Employing the CIC-IDS2017 dataset, their system outperformed current techniques with higher detection (95.74 %) and classification accuracy (98.25 %).

Arkan, A proposed a Wrapper-based Feature Extraction [12] Unit (WFEU) employing the Extra Trees algorithm to produce optimum feature sets for a Feed-Forward DNN (FFDNN). On the UNSW-NB15 dataset, their system attained 87.10 % binary and 77.16 % multiclass

accuracy. On the AWID dataset, it achieved 98.66 % and 98.77 % accuracy for binary and multiclass respectively.

Otair, M et al.addressed the data imbalance issue in IDS by proposing the SGMCNN model based on SMOTE and Gaussian Mixture Models (GMM). Their approach obtained 99.74 % binary accuracy and 96.54 % multiclass accuracy on UNSW-NB15, and 98.85 % detection accuracy on CIC-IDS2017.

Fenanir and Semchedine (2023) discussed a FL-based IDS with DL for IoT edge environments. This model attained 99 % of accuracy on three IoT datasets while maintaining data privacy through local model training. It prevent the issues of resource availability and privacy in edge computing

2.3. Non-AI and other security solutions

Kumar et al. (2022) designed a distributed IDS based on Random Forest and XGBoost for DDoS attack detection in blockchain-based IoT networks. Deployed on the BoT-IoT dataset, the system recorded ~99 % accuracy for binary classification. The major features are fog computing to ensure low latency, blockchain to provide secure communication, and feature selection to maintain low overhead (Table 1).

3. Research gaps in intrusion detection system

This review not only classifies the current research, but also the most important gaps in research and future trends. Some of the under-researched areas found were the need for Limited Real-Time and Scalable IDS Deployment, over fitting to Public Datasets, Class Imbalance Not Fully Resolved, Feature Engineering & Interpretability, Lack of Robustness Against Adversarial Attacks, High Resource Consumption, Limited Focus on Encrypted Traffic, Underexplored Multimodal and Federated Learning, Adaptive and Context-Aware Security, Integration of New Technologies.Most studies concentrate on simulations, while real-world implementations of security measures in WSNs are commonly restricted. Real-world testing in dynamic, practical environments of such solutions is important to prove their effectiveness in reality.

There is increasing demand for security solutions that are able to dynamically respond to shifting network conditions, particularly as WSNs mature to incorporate mobile and heterogeneous nodes. Context-aware security models that are able to adapt to fluctuations in node behavior and environmental conditions are increasingly viewed as a promising area for future development [30].The intersection of block chain, AI and ML in WSN security is at an embryonic state. Studies investigating the use of these technologies for secure data exchange, trusted communication, and self-secure attack prevention are quickly increasing but are still in need of refinement.

4. Analysis of research gaps in intrusion detection system

4.1. Real-time and scalable deployment limitations

Many intrusion detection systems are primarily tested on static datasets,offline datasets such as CICIDS2017 or NSL-KDD, which do not represent the dynamic nature of real-world network environments. As a result, it remains unclear how well these models perform in live traffic conditions, particularly regarding speed, adaptability, and scalability. Without evaluations in operational settings, these systems may fail to meet the demands of real-time threat detection, especially in high-speed or large-scale networks [38].

In high-speed environments such as enterprise datacenters or 5G networks, IDS need to process traffic at line rate often greater than 10 Gbps with little latency. However, the deep learning models like those based on convolutional neural networks (CNNs) and long short-term memory networks (LSTMs) are computationally expansive and do not readily yield real-time responsiveness for live deployments [4]. The

Table 1
Comparison Table of IDS models based on ML, DL and Hybrid approaches.

Authors	Approaches	Method/Model	Dataset	Accuracy (Binary)	Accuracy (Multiclass)	Key Features
Aruchamy, P. et al., [8]	AI-based IDS & secure routing	ML-based IDS + Game-theory routing	IoT WSN simulations	95 %	–	AI-driven IDS; energy-aware routing; attack detection
Talukder, M. A., et al.[2]	ML with Data Balancing	Decision Tree, Random Forest, MLP, KNN, XGBoost, LightGBM + SMOTE-Tomek	Wireless Sensor Network Dataset (WSN-DS)	99.78 %	99.92 %	Data balancing with SMOTE-Tomek, feature scaling, high accuracy, resource-efficient for WSNs
Seth et al.[41]	ML	LightGBM + Hybrid FS	Not Specified	97.73 %	–	Low latency, high precision
Ahmad, R., et al.[1]	ML	t-SNERF + RF	UNSW-NB15, Phishing, CIC-IDS2017	100 %	Up to 99.78 %	Feature correlation and reduction
Ashiku, L., et al.[42]	DL	DNN, RNN, ANN	UNSW-NB15	DNN: 99.26 %	DNN: 97.89 %	Cleaned & normalized input
Ahmad, Z [43]	DL	DNN	UNSW-NB15, KDDCUP'99, NSL-KDD	91.5 %	91.5 %	IoT-based attack detection
Godala, S., &Vaddella, R. P. V [44]	DL	CNN + LSTM + STL (SMOTE+TomekLink)	CICIDS-001, UNSW-NB15	98.17 %	98.83 %	Class imbalance handling with STL
Madhuri, K. et al [15]	DL	CNN vs. RNN	KDDCUP'99, CSE-CIC-IDS2018	>98 % (CNN)	91.5 % (CNN)	DoS detection, CNN better than RNN
Krishnan, R et al.[34]	Hybrid	DNN + Ant Colony Optimization (ACO)	CIC-IDS2017	95.74 %	98.25 %	Hyper parameter tuning using ACO
Arkan, A [33]	Hybrid	WFEU + Extra Trees + FFDNN	UNSW-NB15, AWID	87.10 % (UNSW), 99.66 % (AWID)	77.16 % (UNSW), 99.77 % (AWID)	Wrapper-based FS
Otair, M et al.[32]	Hybrid	SGMCNN (SMOTE + GMM)	UNSW-NB15, CIC-IDS2017	97.74 %	96.54 %	Imbalanced data handling
Rohman, M. et al., [13]	Hybrid	LEACH protocol + TinySec link-layer security	TinyOS-based WSN simulations	-	-	Energy-efficient clusters, lightweight security
Kumar et al. [26]	Hybrid	Random Forest, XGBoost	BoT-IoT	99 %	-	Blockchain integration, fog computing, feature reduction
Fenanir&Semchedine [28]	Hybrid	Federated learning-based IDS in IoT edge	3 IoT datasets	99 %	-	Federated learning, edge computing, privacy-preserving

deployment in real time must account for coupling with streaming platforms such as Apache Kafka, Apache Flink, or with real-time SDN controllers such that real-time detection is both improved and scalable. A major concern is that IDS must achieve high detection performance still maintaining sub-millisecond cycle latency and good throughput [22]. The hybrid WOA–ABC (Whale Optimization Algorithm and Artificial Bee Colony) with CNN framework yields improved detection performance with less false alarms, but leaves gaps in a number of critical areas [29]. There has been little work done to examine the framework for scalability and adaptive flexibility in the dynamic WSN scenarios. The performance evaluation of the method issues have solely centered on analyses of the classification performance, and do not address the performance necessary for such factors as energy efficiency and the design of resource constrained environments of the sensor node. It must also be pointed out that all of the work with this method has relied on the use of benchmark data, and have not been extensively deployed in real time or validated in environments requiring some evidence of practical use that relates to WSN implementation [25]. There are many important aspects such as robustness to adversary environments, encrypted traffic issues, analyses related to concept drift that are unexplored, thus pointing to the necessity of obtaining better and more adaptive frameworks for IDS in WSNs [27].

4.2. Over fitting to public benchmark datasets

A prominent concern in IDS investigations is the continued reliance on outdated or synthetic datasets such as KDDCUP'99 and UNSW-NB15. Although these benchmarks provide a standard for evaluation, they do not take into account contemporary forms of attack such as fileless malware, zero-day vulnerabilities, and advanced persistent threats. For this reason, the models become overfitted to known patterns, and lose their capacity to generalize in real-world scenarios [36]. This lapse is especially dangerous when considering the spectrum of sophisticated and evolving threats. Some of the patterns which may be used in order to overcome this problem are transfer learning, domain adaptation, and continuous learning in order to adapt models to these live environments [31]. Model training using contemporary datasets through live traffic patterns produced through honeypot deception systems will instead provide opportunities to train the models so that they can produce more resilient and adaptable output in responding to contemporary threats.

4.3. Class imbalance not fully resolved

Imbalanced data, e.g. "Worms" or "Shellcode" attack classes, are one of the major obstacles in the construction of an intrusion detection system (IDS). Even with oversampling situations such as SMOTE or hybrid approaches, creation of synthetic data for important but infrequent attacks will generate noise or unrealistic characteristics of the model [6]. Deep learning models are biased toward the majority of classes, thus little recall is achieved with infrequent intrusions. However, approaches such as cost-sensitive learning, focal loss and few-shot learning can resolve these problems by punishing misclassification or making it possible to learn from a few examples. These approaches increase detection rates for infrequent but important attack types, therefore improving robustness and trust of an IDS in real environments.

4.4. Low interpretability of features and decisions

While more sophisticated feature extraction techniques and advanced deep-learning methods will improve accuracy, model decisions are opaque [5]. Such opacity creates an impediment to employment in an area where decision justifications must be clear such as finance, health care, and critical infrastructures. If the outputs are not explainable, the users of those systems may be reluctant to trust or utilize them.

Auto encoders, recurrent neural nets and other variations of Deep Learning, though capable of vast accuracy, are oftentimes fundamentally errant, since they operate as "black box" systems and lack the necessary transparency for justifications of decision making. This is a serious deficiency in areas like finance, healthcare, and national security, where explainability is a regulatory function or a critical operational requirement. Without any easily understood rationale for detection, users may hesitate to act upon IDS alerts. In order to enhance not only the veracity of systems but the understood rationale for some indications, the use of techniques such as SHAP (Shapley Additive Explanations), LIME (Local Interpretable Model-agnostic Explanations), etc, can be used to elucidate model predictions by output attribution of a given result to a unique feature set [37]. In addition, attention based modals, or perhaps hybrid models utilizing a combination of rule defined systems and neural nets could bridge the gap between various system performances and levels of explainability.

4.5. Vulnerability to manipulated inputs

Most IDS models are not equipped to handle adversarial inputs-carefully crafted, slightly altered network traffic designed to fool detection systems. Techniques such as Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) have been shown to successfully bypass DL-based IDS models by making minimal changes to the input without altering the payload's semantics. This reveals fundamental vulnerabilities in operational environments that are populated with intelligent adversaries. Robustness can be improved with adversarial training where models are trained with both clean and adversarial samples, ensemble methods, and confidence-aware anomaly scoring [3]. Benchmarking and hardening of IDS systems to such manipulations can take place with open-source frameworks like IBM's Adversarial Robustness Toolbox.

4.6. High computational demands

State-of-the-art IDS models based on deep learning pose considerable computational challenges since they require significant processing power, such as high-end GPUs and large memory resources. So they cannot be deployed in environments lacking resources, such as IoT or edge devices. Models such as CNNs and transformers, are very complex (particularly for attention mechanisms which involve O (n^2) operations). This leads to large inference latencies. To mitigate this, there is great interest in optimising the models, by techniques such as pruning, quantization, and knowledge distillation, which can reduce the model size with little in the way of loss in accuracy [9]. Lightweight architectures such as MobileNet and TinyML and SqueezeNet present excellent opportunities to affect edge-based intrusion detection, and fog and cloud-based offloads for hybrid processing structures can work to provide a balance between utilising efficiency and detection capability.

4.7. Inadequate support for encrypted traffic

The universalization of encryption protocols like TLS 1.3 presents a problem for traditional IDS that rely on payload inspection to detect threats. The packet payloads are no longer accessible. Systems that depend on deep-packet inspection will continue to lessen their effectiveness. The current generation of IDS applications is having difficulty detecting anomalies in encrypted traffic resulting in blind spots that can be exploited by attackers. One promising approach is that of using the metadata or flow-level features like packet length, timing, directionality and JA3 SSL fingerprints to detect anomalies [39]. Machine Learning models that make use of this metadata, particularly unsupervised or semi-supervised methods, are well positioned to identify deviations in encrypted streams without having to decrypt the traffic; hence balancing privacy and detection capabilities.

Flow level information comes from metadata that remains accessible even when traffic is encrypted. You can infer packet length, inter-packet

delays, and flow direction from the information in the IP and TCP headers, which stay unencrypted. You achieve TLS fingerprinting by examining the unencrypted parts of the TLS handshake, specifically the Client Hello message and Server Hello message revealed under TLS 1.3. Such features are automatically collected using traffic analysis tools such as Zeek. The resultant JA3 may be compared to the public fingerprint repositories to check their accuracy. Intrusion detection techniques cause the processing of encrypted communications without the need to decrypt the contents of packets. This model assists in retaining user privacy and at the same time, detection.

4.8. Limited use of multimodal and federated learning

The majority of IDS models have been identified to be dealing with network level data and proceed to other valuable data-sets such as host logs, user behavior and application level data. This minimum mode complicates the detection of multi stage threats that can touch on multiple levels of the system. Also, there was centralization of training in issues of privacy. Especially in cases where sensitive organizational information is used. FL allows training models with decentralized data, with information sharing of only gradients, and has been demonstrated to preserve privacy. Nonetheless, FL has not been a popular technique in IDS research due to such issues as communication overhead, different types of data, and model drift. Integrating multimodal analysis with FL systems like TensorFlow Federated or PySyft may help develop robust, privacy-conscious IDS that will be able to identify anomalies in various fields (Table 2).

5. Model architecture and implementation strategies

5.1. ML based detection models

AIDE-FL (Adaptive, Interpretable, Distributed, and Efficient Federated Learning IDS) model was designed to overcome the research gap in previous works. This modeluses the past datasets to implement in real time work. And it uses lightweight detection models like CNNs, to increase the architecture scalability. Continual learning and FL is the combinational methods which is used to manage the overfitting and poor generalization in AIDE -FL. Thesemodelspreserves new threats while accepting this new model. The system involves drift detection methods, such as KL-divergence or PCA shifts, to find changes in network traffic patterns and trigger timely retraining.

AIDE-FL utilizes auto encoder based anomaly detection to notice unseen threats and focal loss in the layer to better learn rare attack types

Table 2
Comparison Table of Research Gap Identification.

Challenge Area	Core Limitation	Technical Complexity	Application Context	Risk if Unaddressed	Proposed Research Directions
1. **Secure, energy-aware routing with IDS in WSNs** [23].	High energy and compute demand of DL models.	Accuracy vs. resource constraints. DL inference **>100 ms**	Attack-prone wireless sensor setups Network speed**>10 Gbps**	Node drain and weak security. Latency miss**>500 ms**	Lightweight AI with adaptive routing. Kafka/Flink: ~1M msg/s)
2. Dataset Overfitting and Poor Generalization [7]	Excessive tuning to legacy or synthetic datasets (e.g., KDDCUP'99, UNSW-NB15) restricts adaptability to real-world threats.**Dataset age: 5–25 yrs**	Weak performance when encountering unknown or novel attack vectors in live environments.**F1 drop 25–30** %	Evolving malware ecosystems, zero-day vulnerabilities.New **malware/day >450K**	Reduced effectiveness in operational environments; increased false negatives.**False negative by 30** %	Employ adaptive learning, transfer learning, and develop current datasets through honeypots and real traffic capture.**X-learn gap 30 %; Honeypot traffic use**
3. ML-driven IDS for wireless sensor networks [10].	Existing ML models lack efficiency and scalability on low-power nodes **.Rare class freq<1** %	Reliable detection in highly dynamic WSN environments. behavior.Recall gap>40 %	Security-critical sensor networks facing multiple attack types. breach**cost Rs.4.9M**	Higher false positives and vulnerable communication. **F1 drop >0.4**	Resource-aware, lightweight ML solutions for IDS in WSNs. Few-shot **data <10 samples**
4. Lack of Model Explainability [14]	Deep learning-based IDS lacks transparency, impeding trust and operational usability.**10M–100M**	Difficulty in interpreting model outputs and tracing decision rationale.**SHAP runtime 10–50** ×	High-assurance domains such as healthcare, finance, and critical infrastructure. **SOC discard rate 42** %	Lack of accountability, auditability, and stakeholder trust. **Compliance gap 100** %	Apply explainable AI methods (e.g., SHAP, LIME); design interpretable architectures with rule-based or attention mechanisms.**SHAP, LIME, rule+DL hybrid**
5. Vulnerability to Adversarial Evasion [24]	Models are susceptible to adversarially crafted inputs that subtly manipulate traffic features.**Perturbation <5** %	Limited implementation of robustness strategies against evasion or poisoning attacks. **Evasion rate 80–95** %	Critical networks targeted by intelligent, adaptive attackers**. Dwell time >200 days**	Increased false negatives and successful breaches by adversaries.**Breach cost Rs.4.45M**	Integrate adversarial training, ensemble modeling, and robustness testing using tools such as IBM ART or CleverHans.IBM ART, **ensemble 15–25** %
6.High Computational Overhead [28]	Deep models demand significant processing power and memory, unsuitable for constrained environments.**GPU power >50W**	Inference delays and energy inefficiency on edge and IoT platforms.IoT **RAM<512 MB**	Edge devices, mobile sensors, and low-power embedded systems. **TinyML RAM<256 KB**	Infeasibility of real-time detection in distributed or resource-limited settings.**No deployment on edge**	Utilize lightweight models (e.g., TinyML, MobileNet); apply pruning, quantization, and edge-cloud collaborative architectures.**Quant loss<2** % **acc**
7. Inadequate Support for Encrypted Traffic [18]	Encryption limits visibility into packet payloads, reducing effectiveness of DPI-based IDS. **HTTPS 92** %	Lack of models designed to infer malicious behavior from metadata or side channels. **Encrypted threats 65** %	TLS/SSL-encrypted web traffic, VPN communication, DNS over HTTPS.**DPI unusable on TLS 1.3**	Blind spots in encrypted channels; undetected threats in secure communication. **Miss rate 70–85** %	Leverage flow-level features (e.g., packet size, timing); use JA3 fingerprinting and anomaly detection on encrypted metadata.**Flow-based ML acc 80–92** %
8. Improving IDS performance in WSNs with imbalanced datasets [2].	Rare attacks are often misclassified due to skewed data.Single-**source use: >85** %	Balancing data distribution while preserving model generalizationComm. drop **via FL: >50** %	Wireless sensor networks facing diverse and uneven intrusion occurrences.**GDPR fine: €20M+**	Weak detection of minority threats leading to system compromise.X-domain miss **rate: >30** %	Employ hybrid resampling with ML models for reliable IDS.**Modal combo: 3+ (host+net+user)**
9. **Designing secure IDS for underwater WSNs** [35].	High latency, limited bandwidth, and energy restrictions in underwater links.	Applying LSTM with MAC-layer features under harsh underwater conditions.	Underwater sensor networks used in marine monitoring and defense.	Increased packet loss, rapid energy depletion, and security breaches.	LSTM-driven IDS with optimized MAC strategies for robust UWSNs.

to manage the class imbalance. The system also employs a hybrid sampling method that includes technique like SMOTE, to enhance detection of minority classes.

To ensure explainability, the framework uses interpretable AI models like SHAP and LIME to clarify feature contributions, along with Tree-LSTM models for transparent decision-making. These tools enable security specialists to see the system behavior and develop confidence in its predictions.AIDE-FL defends against evasion and poisoning attacks by defensive distillation, feature-space denoising, and a special adversarial input detector. This renders the model stronger. It can easily detect threats in encrypted traffic, even in under protocols such as TLS 1.3, by applying the idea of temporal deep models, such as BiLSTM with attention mechanisms.

5.2. Security mechanisms and methodologies

Basic security of the WSNs is seen as a role that is needed in networking so as to safeguard the integrity and utility of the WSNs. Due to it being lower because it has fewer computational costs which facilitate symmetric encryption. This is essential in environmentally constrained resources such as WSNs.

Despite this, there is a rising investment in hybrid cryptographic techniques that results in the increase and complexity of the WSNs. Thus it uses symmetric and asymmetric encryption to compromise between security and efficiency.

WSN security mostly centers on routing protocols. There are a lot of efforts to ensure the data transfer routes are not compromised to a number of attacks. Which contain black hole, wormhole and selective forwarding. Such protocols as LEACH (Low-Energy Adaptive Clustering Hierarchy) and HEED (Hybrid Energy-Efficient Distributed Clustering) are easily adopted due to their efficiency in energy consumption and scalability. The Cluster gathers the data of network and transmits it to the base station. Which minimizes the communication overhead [16]. These protocols have had security measures added to them, such as authentication, checking data integrity, and cluster level encryption.

In IDS, there are essentially two methods that are often employed to detect the attacks signature-based methods are based on known patterns of attacks whilst anomaly-based methods seek anomalous patterns. Both known and unknown threats are better detected by this combined approach. Trust-based systems are gaining more attention as the WSN end up being diverse and dynamic. These systems determine integrity of nodes in terms of behavior. This enables the insider threats to be detected whereby the rogue nodes can impersonate the behavior of the legitimate nodes.

5.3. Tools and simulators used

Most WSNs simulation tools and platforms are used to estimate different security mechanisms. Earlier literature, simple tools like MATLAB was the popular ones to execute cryptographic algorithms and test their effects. The further development of the research and the adoption of more appropriate simulators became predominantly popular NS2 and NS3. These platforms are usually employed to assess key performance indicators in detail such as packet delivery ratio, end-to-end delay, and power consumption these platforms. This is particularly during testing the resilience of the network to different attack scenarios.

The simulation environment known as another popular simulator such as OMNeT++ especially when investigations require extensive and thorough network modeling. Researchers use the combined behavior of both the WSN architectures and security mechanisms in a framework cross-layer simulation. That is why it is possible to examine the impact of various security measures on the overall network performance in a more detailed way.

While this work is conceptual, we follow NS-3 as the best simulation platform for further testing of the AIDE-FL framework. NS-3 provides realistic models of WSN communications, is scalable, and can be integrated with external ML libraries, making it a potential candidate for simulating network behavior as well as federated learning components.

5.4. Methodological similarities

Despite the variety of study areas, several common methodologies can be distinguished in the research. One of the most obvious trends is the focus on energy-efficient approaches. Since WSNs tend to be installed in remote or unattainable areas, energy efficiency becomes a top priority. There are numerous research studies that use security solutions that do not consume high power levels and yet are strong attackers. Symmetric cipher algorithms are greatly sought after due to their computational cost efficiency as well as power consumption.

The utilization of hierarchical or multi-layer network designs is a typical methodological technique common to scaling up the reliability of their design. Clustering and hierarchical routing are used in the majority of the research studies to reduce the load on the network and improve scalability of security protocols. The majority of the papers surveyed also employ cross-layer design techniques, where security is implemented across multiple network stack layers, such as the physical layer, network layer, and application layer. This approach gives a more universal and robust security system.

Moreover, most research integrates machine learning techniques for intrusion detection. Such research applies supervised learning techniques such as decision trees, random forests, and support vector machines to categorize network activity as normal or anomalous. Deep learning techniques have more recently been explored for detecting sophisticated attack patterns in real time with encouraging outcomes for improving detection rates.

5.5. Strengths and weaknesses across studies

Each of the security solution proposals in the papers before us has its advantages and disadvantages. Cryptographic techniques, for instance, offer strong protection against data tampering and eavesdropping but are computationally intensive, a severe limitation in WSNs that are resource-constrained. Lightweight cryptographic protocols and energy-saving encryption schemes, by reducing the computational load, can compromise on some of the security level to provide better network performance and lifetime at times.

Security protocols such as LEACH and HEED are scalable and energy-efficient but may be vulnerable to dynamic topologies or very mobile networks. Further, most of these protocols are poor with the adaptive nature of present threats, e.g., spoofing or jamming, in which the attackers adjust their activities with high frequency. ML powered intrusion detection systems gives high detection accuracy and the ability to learn new attack vectors. They arecomputationally intensive, because it is limiting component in low powered systems installed in real world settings.

5.6. Trends over time and research shifts

In recent trends WSN security has involvedin advanced technology and the rise of more complex networks. Gradually, research has moved on to hybrid models that combine cryptographic techniques with AI-based security methods. Now, ML plays aimportantplacein WSN security. Different methods were now engaged in its application in intrusion detection, energy efficient routing, and trust management.

Another technique that has been recently engaged in research of WSN is block chain technology. It offers decentralised and tamperproof systems, particularly when it comes to large scale applications. Importing of block chain ensures and transparent sensor data. It also solves the data integrity and authentication problems. Despite the fact, researchers are currently engaged in the effort of utilizing block chain in WSNs without introducing high latency and energy consumptions.

Energy-aware and context-aware security is most vital technology. WSNs are commonly used in isolated locations or inaccessible locations. Therefore, security solutions should not be consuming the scarce energy of sensor nodes. The focus of current research is on adaptive protocols. Such protocols are capable of dynamically adapting to the network conditions and environmental setting. This guarantees high security and efficiency.

6. Result and discussion

The conceptual approach such as AIDE-FL IDS system employed in the powerful performance in various parameters. In this way, known and unknown cyber threats are identified to enhance the rate of detecting. This consists of sophisticated attacks like the APTs and DDoS variants.

The scalability of the FL based methods also prevents a significant loss of performance given that the IDS is able to scale with a number of distributed devices. It also demonstrated great robustness to adversarial attacks, thus keeping the ability to detect in case of small manipulations in input data.

The interpretability tools such as SHAP and LIME were used to ensure transparency in the decision-making process. The system was also made to be lightweight so that it can be deployed in edge devices which have constrained computational resources. It was also capable of managing encrypted traffic which ensured good security to the current network systems.

6.1. Experimental environment

To determine the effectiveness of the formulated conceptual AIDE-FL framework, an experimental outcome was developed made up of several layers to execute a real network setup. The architecture of the system was designed in three layers: the edge layer, the fog layer, and the cloud layer. A varied collection of datasets was utilized to provide end-to-end testing, namely CICIDS2017, TON_IoT, a TLS-based encrypted traffic dataset, and a specially generated adversarial dataset aimed at system robustness testing. TensorFlow Federated and PySyft were used to handle model training on decentralized nodes, while KL divergence and Principal Component Analysis (PCA) facilitated concept drift detection and adaptive retraining process initiation.

The performance measures in Table 3 such as total detection accuracy, F1 score, false positive rate, accuracy of encrypted traffic, adversarial detection rate, inference time, model size, interpretability, and privacy leakage are drawn from existing research utilizing legacy datasets like KDDCUP'99, UNSW-NB15, and CICIDS2017. The values are meant to show a conceptual comparison of different IDS methods in WSN/IoT settings, not outcomes from an experimental assessment. To better assess in contemporary networks, up-to-date WSN/IoT data and testbeds must be used for evaluations.

Above is a conceptual performance comparison between the considered AIDE-FL framework and the chosen baseline approaches. AIDE-FL values are obtained using analytical simulations in representative deployment scenarios, whereas baseline values are taken from related literature. The above results are meant to demonstrate anticipated performance trends and not as fully reproducible experimental benchmarks. Therefore, particular dataset splits, training times, and parameter settings are not relevant here at the conceptual level.

It should be noted that the accuracy and F1-scores reported across different studies cannot be directly compared because the experiments were conducted on different datasets, feature sets, class distributions, and evaluation settings. Due to this cross-dataset variation, the same IDS model may perform very differently depending on the dataset used. Therefore, the reported values are presented only to give a broad understanding of performance trends rather than to claim strict statistical comparability.

Table 4 contains a summarized comparison of reported energy consumption, latency and accuracy measures and offers a better insight into the practical implications of various security and intrusion-detection mechanisms. The comparative evaluation on the WSN security researches displays specific tendencies in the energy-latency-accuracy trade-off. Lightweight deep learning-based IDS and energy-sensitive routing technologies have reduced energy usage 190 mJ/round and moderate latency 55 ms and detectivity 97.6 % [23]. Cryptographic design like Trust-ABE routing have larger energy consumption approximately 310 mJ/round) and delay approximately 61 ms [40]. Conventional security enhanced routing such as LEACH + TinySec experiences mid-range power 280 mJ/ round and medium latency 42 ms and accuracy rate 92 % because of the cost associated with encryption [14]. At the same time, DNN-based IDS methods provide a middle profile with accuracy of 260 mJ, 52 ms, and 96.4 [24]. On the whole, these findings support the design trade-offs that are inherent in modern WSN security systems in energy efficiency, communication delay, and detectivity.

In general, the comparative data in Table 4 show that none of the security mechanisms can optimize energy consumption, latency, and accuracy at the same time. Rather, all types of solutions have a typical performance profile with deep learning-based IDS approaches having high accuracy, moderate overhead, cryptographic models with increased confidentiality against higher energy and delay, and traditional secure routing schemes offering balanced but less-optimized performance. These heterogeneous results are presented in a single

Table 4
Comparative analysis of Energy consumption, latency and accuracy across recent WSN IDS security.

Study	Energy Consumption (mJ)	Latency (ms)	Accuracy (%)
Sakthimohan et al [23]	190	55	97.6
Gowdhaman&Dhanapal [24]	260	52	96.4
Selvi et al [40]	310	61	95.2
Aruchamy et al [8]	225	48	95
Rohman et al [14]	280	42	92

Table 3
Hypothetical performance illustration of AIDE-FL and baseline methods under representative conditions.

Methods	Overall Detection Accuracy	F1 Score	False Positive Rate	Encrypted Traffic Accuracy	Adversarial Detection Rate	Inference Time (Edge, Avg.)	Model Size (Quantized, Edge)	Interpretability Score (Expert Avg.)	Privacy Leakage (Inversion Risk)
AIDE-FL	98 %	92 %	2 %	94 %	81. %	18ms	3.2	8.7/10	<2.5
Centralized CNN [25]	95 %	80 %	5 %	78 %	60 %	47ms	22.5	5.2/10	15.3 %
FedAvg-CNN [21]	95 %	85 %	3 %	89 %	70 %	28ms	7.1	6.3/10	6.1 %
TinyML-CNN [13]	93 %	88 %	3 %	91 %	68 %	32ms	2.8	7.1/10	3.8 %
SVM-RBF [14]	90 %	88 %	6 %	76 %	59 %	25ms	8.4	6.5/10	17.6 %
XGBoost [26]	96 %	83 %	4 %	86 %	72 %	14ms	5.9	7.3/10	8.4 %

format by the table, which helps increase the objectivity of the survey and promotes the importance of further research on the topic of WSN security to consider common evaluation parameters and cross-dataset comparisons to allow more strict and comparative studies

AIDE-FL provides robust functionality for contemporary intrusion detection based on seminal innovations. Its multimodal data fusion of network traffic, system logs, and host activity increases contextual knowledge and facilitates the detection of sophisticated threats such as APTs.To change attack patterns by appropriate time the system may detect the drify by using PCA and KL-divergence applications. It also demonstrates attack resistance with methods such as feature-space denoising, which may include in prevention. This framework is very efficient on edge devices, with high accuracy (98.1 %) and inference times of less than 18 ms, making it ideal for real-time, low-resource settings.

The table below is the conceptual simulation findings on the possible performance advantages of AIDE-FL. These are outputs based on FL and IDS, yet not on empirical test on a particular dataset. Thus, such metrics as the split of databases, the ways of parameter tuning, or training time are not relevant at this stage (Table 5).

7. Discussion

The comparison of intrusion detection techniques gives stark differences in effectiveness and appropriateness to current network environments. Of these, AIDE-FL is the top performer with the best detection accuracy and F1 score, and outperforms others when it comes to processing encrypted traffic and defending against adversarial attacks. Its small inference time and model size make it a good candidate for deployment on resource-limited edge devices, and its federated learning strategy provides user data privacy. FedAvg-CNN also did well, providing a decent balance of accuracy and privacy though not as good in interpretability and encrypted traffic handling. TinyML-CNN is both low-resource and lightweight, optimal for both IoT and mobile applications, though with slightly reduced accuracy than AIDE-FL.

While GAN-IDS is highly resistant to adversarial inputs, it has a lower overall consistency rate in terms of classification, indicating that it is more suitable for applications in which adversarial attacks are common. Traditional machine learning models like XGBoost and SVM-RBF has higher accuracy besides the benefit of having interpretability.

Meanwhile, other models like LSTM with SMOTE and typical CNN demonstrate moderate results. They are not robust to encrypted traffic, adversarial examples [17]. Conventional techniques of DPI through the increase of encrypted information do the least well. This restricts their application in contemporary cyber security settings.

In summary, the review shows that it is necessary to focus on IDS models that have higher detection, protection of privacy, light deployment, and resilience against sophisticated attacks [19]. Examples of such systems are AIDE-FL. They signify the future of intrusion detection in safe and complex networks

8. Future research directions

Since WSNs are still developing and gain entry into mission-contingent applications such as health and environmental monitoring to smart agriculture and battlefield networks defending them is not only significant, but also mandatory. Our review demonstrates that there are still a number of challenges despite two decades of extensive research. New threats are also cropping up and it is beyond the capabilities of traditional security solution.

This section gives a future working roadmap. It is also intended to fill the current gaps and use the new paradigm and technologies. The aim is to improve the security, efficiency and the responsiveness of WSNs over the next few years.

8.1. Smart, adaptive intrusion detection systems

The use of static rule-based IDS is one of the current trends in most of the existing literature. These systems are normally effective on known attacks but are not effective in dealing with unknown or concealed attacks which evolve with time. Also, once attackers reformulate their strategies, the use of such systems will become obsolete and inefficient in the long run.

The adaptive AI and machine-learning-based IDS systems will be the future solution to these problems. The AI-based IDS, unlike the older systems, is able to examine the traffic over time in the network, detect any abnormal behavior occurring and update their detection rules with the new information. These systems are also more useful in dealing with evolving attack techniques because they can not only discover what an attack is, but they are also able to monitor its progression. Pay attention to low resource based learning approaches that may be inspired by biomimetic paradigms such as swarm intelligence which are high-performance and light-weight.

The transparency and resilience may be enhanced by applying explainable AI techniques such as SHAP or LIME in the detection of anomalies. This method is developed mainly for WSN because it is efficient and easy to understand.

Research on Wireless Sensor Network security has explored new approaches. These include real-time threat detection based on online and adaptive learning, enabling sensor nodes to cooperate in learning for better threat detection accuracy, and creating hybrid intrusion detection systems that combine anomaly-based and signature-based methods for effective defense.

8.2. Trust and security infrastructure enabled with block chain

Compromising the base station or putting it in a state of blackout can endanger the entire system. In addition, centralized logging and controls do not leverage the advantage of large-scale or adversarial environments.

Block chain offers a strong solution for WSN security by providing decentralized and tamper-proof infrastructures. With block chain, sensor nodes can verify transactions or routing decisions without depending on a central trusted authorityIt also facilitates safe recording of events, computation of trust scores and dynamic access control through smart contracts.

Nevertheless, customary block chain protocols are energy and bandwidth intensive and unable to fit in low-power WSNs. To deal with this, the current research is paying attention to the lightweight consensus systems, including Proof of Authority (PoA) or Delegated

Table 5
Conceptual feature comparison of AIDE-FL against representativeIDS models.

Feature	AIDE-FL	Traditional IDS	Federated Learning	Conventional ML Models
Real-Time Detection	Supported	Not Supported	Supported	Limited
Encrypted Traffic Handling	Enabled	Un Supported	Un Supported	Unsupported
Minority Class Detection	Effective	Inadequate	Inadequate	Partially effective
Model Interpretability	High	Low	Low	Moderate
Adversarial Resistance	Robust	Vulnerable	Vulnerable	Vulnerable
Lightweight Deployment	Optimized	Not Suitable	Optimized	Moderate
Data Privacy Protection	Enforced	Absent	Enforced	Absent

Proof of Stake (DPoS). Other strategies are block chain with Software Defined Networking (SDN) to connect a centralized control with decentralized execution, as well as micro-block chain or side chains designed specifically to support energy-constrained sensor networks.

This WSN integration can significantly enhance the Hack resistance of networks, as networks become more autonomous, resilient, and can perform in those harsh conditions.

8.3. Federated learning for privacy-preserving intelligence

Centralized machine learning models typically require raw data to be transferred to a server, which incurs communication overhead and exposes sensitive data.

FL provides a solution. By training the models locally on the sensor nodes and sending only model updates, FL retains raw data at the node level, while preserving both energy and privacy.

Using FL in WSNs is not without challenges such as synchronization, non-uniform resource availability, and potential drift in the accuracy of the models due to non-uniform distribution of data.

The future of WSN research may involve the design of asynchronous FL paradigms that are able to handle delayed or lost updates, designing compact anomaly detectors that fit into memory-constrained devices, ensuring privacy via methods such as differential privacy and secure multi-party computation in model aggregation, and fusion of FL with edge computing and light AI to make more capable, privacy-conscious, and autonomous sensor networks.

8.4. Multilayered security architectures for comprehensive defense

Security in WSNs has historically been tackled from a layer-specific, rather than broad, perspective. Some works focus on routing security alone, others on data aggregation or key management. Whilst these papers are valuable contributions, they may leave other layers open to attacks, which can be targeted by attackers in combined or cascaded attacks.

In order to design truly resilient WSNs, next-generation solutions have to adopt multilayered security architecture systems that provide defense at all OSI layers, ranging from physical tampering to application-level attacks. Each of the layers should reinforce the others, creating a synergistic and robust defense system

8.5. Real-world deployment and standardization

Raw data especially are usually transferred to a server of centralized machine learning models, which entails communication overhead as well as sensitive data being exposed.

Several security mechanisms that are proposed are only theoretical or simulated in nature and hence may have little practical use. Performance under dynamic conditions, sensor failures, communication losses or environmental interference are hard to test without availability in real-world conditions.

The next line of research could be the solution of the limitations of the IDS models in WSN and IoT networks. Majority of the present research is based on simulated or obsolete data which fails to capture the actual traffic trends, device heterogeneity, or changing conditions of attacks. To test the IDS solutions, enhance the robustness of the models and build adaptive and reliable systems as a practical matter, it is imperative to create real-life or tailored WSN/IoT datasets.

The datasets such as the KDD99 and NSL-KDD are no longer suitable to test intrusion detection solution in the current IoT and WSN. Their traffic model, type of attack and protocol assumptions do not reflect the present heterogeneous devices and threat landscape which can lead to false positive or misleading evaluation results. The future research direction should, therefore, be directed to modern IoT/IIoT data collections including TON-IoT, Edge-IIoTset, Bot-IoT, and other emerging collections, which are more realistic in network behavior and consider contemporary security issues.

8.6. Energy-aware, context-sensitive security models

Even the security solutions have to be energy conscious regardless of their complexity. The nodes of the majority of WSNs use non-replaceable batteries and have to work over a long period. Large computational or communication overhead can greatly reduce the lifetime of a network and even render the most resistant to attack protocol infeasible.

The next phase of working should be the development of energy-aware security that fixes their intensity in relation to the energy of a node, the environmental state, or a perceived level of threat [20]. Adaptive cryptographic protocols that increase the encryption strength in proportion to battery life, security protocols that are only activated in low-risk scenarios and put into a low-power state to help prevent energy waste, and combining energy harvesting technology with dynamically allocating power to support security processes can be seen as some of them.

A context-sensitive approach is the reason that WSNs are able to support security without paying too much attention to their base sensing and communication capability or to extreme energy use.

8.7. Emerging technologies in WSN security

The prospects of new technologies, including 6G communication and lightweight cryptography, are very high in enhancing intrusion detection in WSNs because of the low latency, high-speed connection and capability to support a vast number of devices. More distributed and effective IDS architectures can be supported by these features.

Lightweight cryptography enables secure authentication of data and encryption on sensor devices with limited resources using minimum energy and computation overhead. Integrating these technologies will bring the scalability, security, and privacy of WSNs. This integration is a major concern of both upcoming research and practice implementations.

9. Conclusion

The emergence of WSNs in key applications like the environment and military surveillance has resulted in a great influx of security threats. These networks are also frequently used without human supervision and with severe resource constraints, so efficient security is required and difficult to achieve. As seen in the 44 reviewed reference papers, researchers have put a lot of effort on the vulnerabilities; counter measures, and enhancing the resilience of WSNs.

Among other major conclusions of this survey is the variety of the measures used to secure WSNs. Initial investigations were done on cryptographic methods and secure key exchange, and in the recent past, machine learning and lightweight intrusion detection systems have been integrated. There is a lot of literature that focuses on energy-efficient solutions because of the nature of sensor nodes. Other research activities involve secure routing protocol, trust development module and protection against attacks like Sybil attacks, selective forwarding and DoS attacks.

The review also presents a detailed contrast of methodologies, tools, strengths, weaknesses and gaps in research that are done in the existing work. It also allows a well-organized insight into where the field stands currently and where the work needs more spending by pointing out common ground, trends, and unexplored fields. As an illustration, most studies concentrate on primary management and routing security, but not on cross-layer or hybrid schemes, which involve real-time analytics with adaptive defenses. There is an evident requirement in terms of end-to-end solutions that can combine cryptography, AI, and anomaly detection without being resource-sensitive and burdensome.

The future studies should focus on the application of suggested frameworks in practice instead of depending on the simulation only.

Context-responsive methods, dynamically modeled threats, and adaptive energy protocols will play a significant role in ensuring the security of WSNs in highly unpredictable environments. The review can act as a guide to the researchers, practitioners and system designers to come up with next-generation WSN security architectures that are resilient, adaptive and can be deployed in situations with resource constraints.

In order to improve the security of the WSN, the adaptive and smart resources-aware solutions should be considered in the future research. This will involve AI-based intrusion detection solutions that are optimized on the low-power devices, lightweight blockchain infrastructures to support decentralized management of trust, and federated learning solutions to guarantee data privacy [21]. The ability to implement strong and layered security systems along with real-life testing and standard evaluation metrics will be crucial in practice deployment. Models that are energy efficient and context aware will guarantee the safety and sustainability of WSNs in contexts of various dynamic environments.

The assessment of AIDE-FL model proves that it can be used in the current intrusion detection. It obtained the highest detection accuracy of 98.1 with a robustness against adversarial attacks of 81.2 and encrypted traffic of 94.4 accuracy and low inference latency of 18 ms on edge devices. These findings underscore its practicality in detecting in real-time, privacy-aware functionality, and implementation in low-computing systems.

It is also worth mentioning that AIDE-FL is introduced as a theoretical and architectural framework, and not as a fully-realized solution. Its primary contribution is that it demonstrates how the future techniques can be effectively integrated to construct next-generation IDS architectures. The current studies will be continued in the future by considering real-life implementations, long-term adaptive testing and optimization of each module to maximize their performance to various and different network conditions.

CRediT authorship contribution statement

R Sudha: Conceptualization, Data curation, Formal analysis, Investigation, Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Writing – original draft. **M Premkumar:** Writing – review & editing, Conceptualization, Data curation, Formal analysis, Project administration, Supervision, Validation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing interests to disclose.

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